From Story to Music: Differential Geometry for Emotion-Driven Music Synthesis

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Abstract

This paper presents a pipeline to convert textual stories into expressive music using emotion extraction, differential geometry, and algorithmic composition. Valence and arousal curves are mapped to musical parameters through Frenet frames, curvature, torsion, and Spiral Array tonal mapping.

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1 Introduction

Music generation from text combines NLP, affective computing, and algorithmic composition. We use differential geometry to model musical expressivity: curvature, torsion, and Frenet frames guide pitch, velocity, and rhythm.

Our system has six modules:

1. `core_emotion.py` – Extracts Valence (V) and Arousal (A) from sentences.
2. `geometry_engine.py` – Builds splines and computes Frenet frame quantities: curvature, torsion, tangent, normal, binormal vectors.
3. `role_engine.py` – Maps geometry curves to musical roles, generating pitch, velocity, and rhythm sequences.
4. `composer.py` – Converts role curves into MIDI using scales, chords, Spiral Array mapping, and voice leading.
5. `main.py` – GUI, orchestration, WAV rendering, and mixing.
6. `role_curves_output_images.py` – Generates and saves visualizations of V/A curves, curvature/torsion, and role curves.

2 Related Work

- Tymoczko (2006) – Geometry of chords and voice-leading spaces.
- Chew (2000) – Spiral Array Model for tonal distance.
- Toussaint (2013) – Rhythmic geometry.
- Dorffner et al. (2019), Barbero et al. (2021) – Emotion curves mapped to musical features.
- Gul (2023) – Musical features as smooth parametric curves.

3 Methodology

3.1 Emotion Extraction

Text is split into sentences, each producing Valence (V) and Arousal (A):

```
V_points, A_points = [], []
for s in sentences:
    _, v, a, _ = extractor.get_VA(s)
    V_points.append(v)
    A_points.append(a)
```

3.2 Differential Geometry Curves

3.2.1 Parametric Curve Construction

V/A points are mapped to a 3D parametric curve:

$$\mathbf{r}(t) = \begin{bmatrix} t \\ V(t) \\ A(t) \end{bmatrix}, \quad t \in [0, T]$$

```
v_spline, a_spline = build_splines_from_points(V_points,
    A_points)
T, V, A = sample_splines(v_spline, a_spline,
   duration_sec=duration, samples_per_sec=sample_per_sec
   )
```

3.2.2 Frenet Frame

The Frenet frame defines tangent, normal, and binormal:

$$\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{\|\mathbf{r}'(t)\|}, \quad \mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|}, \quad \mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t)$$

Curvature and torsion:

$$\kappa(t) = \frac{\|\mathbf{r}'(t) \times \mathbf{r}''(t)\|}{\|\mathbf{r}'(t)\|^3}, \quad \tau(t) = \frac{(\mathbf{r}'(t) \times \mathbf{r}''(t)) \cdot \mathbf{r}'''(t)}{\|\mathbf{r}'(t) \times \mathbf{r}''(t)\|^2}$$

```
K, Tau, Theta = compute_frenet_quantities(v_spline,
   a_spline, T)
```

3.3 Role Curve Mapping

Instrument roles map curves to musical parameters:

```
for role, data in role_curves.items():
    V = smooth_curve(data["V"], 3)
    A = smooth_curve(data["A"], 3)
    Theta = smooth_curve(data["Theta"], 3)
```

Pitch mapping:

$$\text{pitch}_{role}(t) = w_V V(t) + w_\kappa \kappa(t) + w_\tau \tau(t) + w_\theta \theta(t)$$

Velocity mapping:

$$\text{velocity}_{role}(t) = 0.6\tau(t) + 0.4A(t)$$

3.4 Harmonic and Tonal Mapping

Spiral Array for tonal distance:

$$d(p, c) = \text{mean}_{c_i \in \text{chord}} \|\mathbf{pc}(p) - \mathbf{pc}(c_i)\|$$

```
pitches_quantized = quantize_to_scale(float_pitches,
   scale_notes)
pitches_chord_aligned = np.array([nearest_chord_tone(p,
   chord) for p in pitches_quantized])
```

3.5 MIDI Generation

```
midi_obj = build_midi_for_role(
    pitches, velocities, duration=self.duration_sec,
    program=presets_map[role]['prog'], role_name=role
)
```

4 Output Visualization

4.1 All Generated Curves

All images are assumed to be in the same folder as the ‘.tex‘ file.

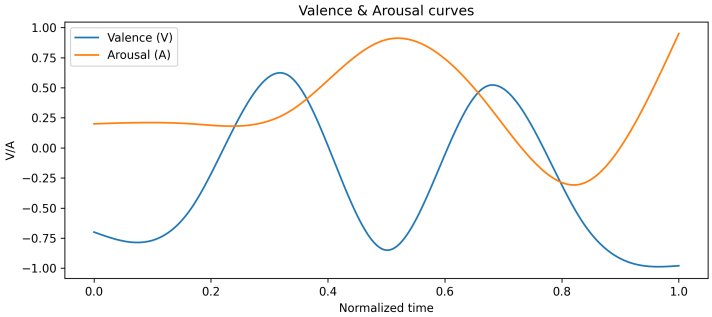


Figure 1: Valence and Arousal curve.

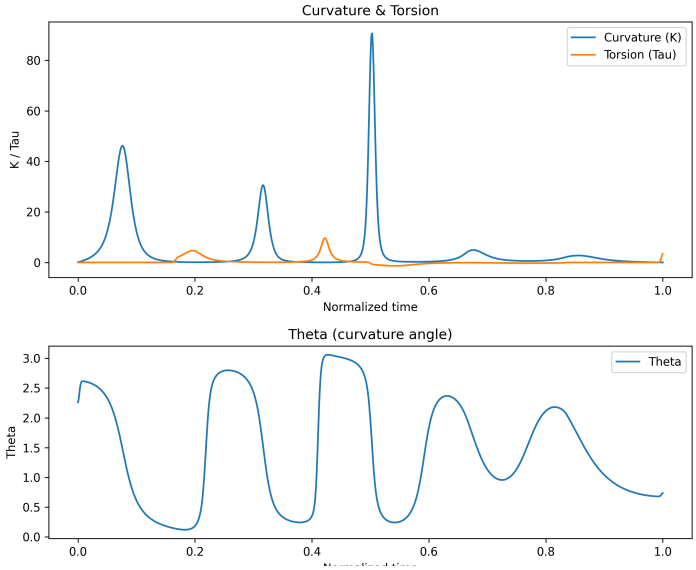


Figure 2: Curvature, Torsion, and Theta over time.

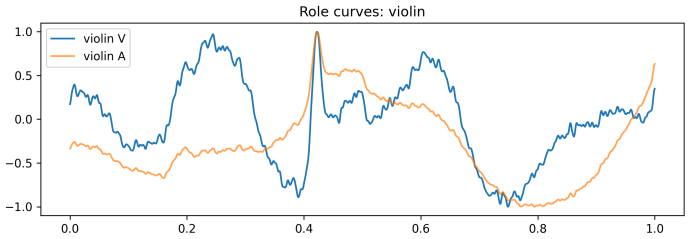


Figure 3: Violin role curve mapping.

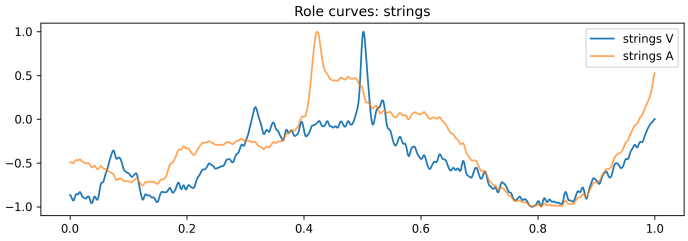


Figure 4: String role curve mapping.

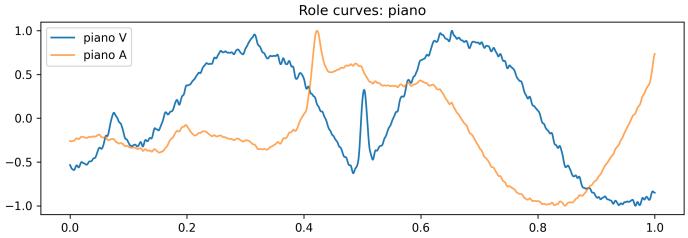


Figure 5: Piano role curve mapping.

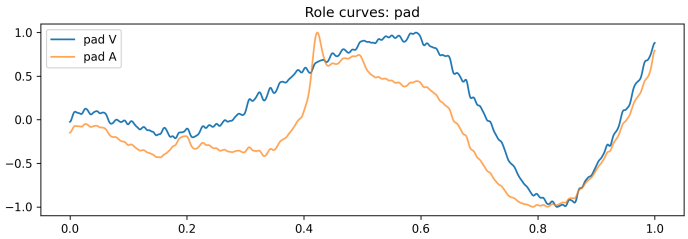


Figure 6: Pad role curve mapping.

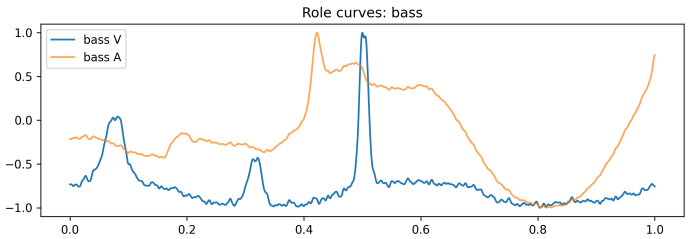


Figure 7: Bass role curve mapping.

5 Pipeline Execution

GUI orchestrates:

1. Story input → V/A extraction

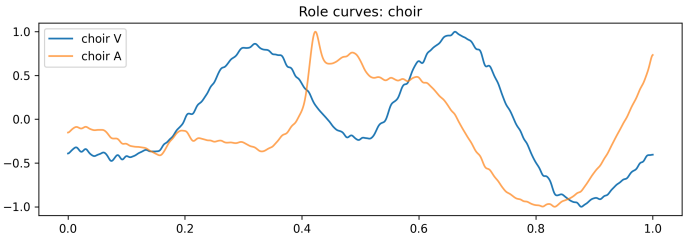


Figure 8: Choir role curve mapping.

2. Spline generation → Frenet quantities
3. Role curves → MIDI
4. WAV rendering → Mixing
5. Curve image saving

6 Discussion and Future Work

- Differential geometry provides **smooth control** of musical parameters.
- Spiral Array ensures **tonal coherence**. - Future Work:

- Incorporate **Iranian traditional instruments** (santur, tar, kamancheh).
- Design **emotion-specific music profiles** for each basic emotion.
- Multi-scale curvature analysis for polyphony.
- Real-time interactive music generation.

7 References

1. Gul, S. (2023). *Music through Curve Insights.* Mathematics 11(20), 4398.
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7. Barbero et al. (2021). *Emotional Curves and Music Generation*, IEEE Trans. on Affective Computing.